

Rescaling-Aware Training for Efficient Deployment of Deep Learning Models on Full-Integer Hardware

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Abstract—Integer AI inference significantly reduces computational complexity in embedded systems. Quantization-aware training (QAT) helps mitigate accuracy degradation associated with post-training quantization but still overlooks the impact of integer rescaling during inference, which is a hardware costly operation in integer-only AI inference. This work shows that rescaling cost can be dramatically reduced post-training, by applying a stronger quantization to the rescale multiplicands at no model-quality loss. Furthermore, we introduce Rescale-Aware Training, a fine tuning method for ultra-low bit-width rescaling multiplicands. Experiments show that even with $8\times$ reduced rescaler widths, the full accuracy is preserved through minimal incremental retraining. This enables more energy-efficient and cost-efficient AI inference for resource-constrained embedded systems.

Index Terms—Quantization, Edge AI, Hardware Accelerator

I. INTRODUCTION

Hardware-efficient full-integer-quantized AI inference is the de-facto standard in today's embedded systems. A crucial step of full-integer quantization is the rescaling of high bit-width accumulators to low bitwidth outputs. This operation is implemented in integer-only hardware by multiplying the 32-bit accumulator with a wide unsigned integer, m , followed by a shift-right-and-round and casting to 8-bit [1]. Consequently, commercial embedded integer neural processing units (NPUs) such as [2] integrate multiple wide multipliers for data-parallel processing (see Fig. 1)—causing significant hardware cost.

One approach to reduce rescaling overhead is to restrict the rescaler to a power-of-two, allowing the operation to be implemented using simple bit-shifting [3]–[6]. However, this requires coarse discretization of the quantization scales, which can lead to noticeable degradation in model accuracy.

We aim to optimize the cost-quality trade-off by reducing the bit-width of m from 32 bits to a smaller value. This also decreases the width of the multiplication output, reducing the cost of the subsequent shift-and-cast operation. We show that one can minimize the rescaling width to 8 bit, from today's 32 bit, without impairing model accuracy.

To enable even lower bitwidths for m , we propose a fine-tuning method called *Rescale-Aware Training*. This method emulates the quantization errors of the rescale hardware during the forward pass. Proper gradient propagation during the backward pass is ensured by Straight-Through Estimation. By leveraging the inherent noise resilience of deep neural networks, our approach enables an $8\times$ reduction in the bit-width of the rescaler multiplicand while preserving model accuracy, all within a reasonable retraining effort.

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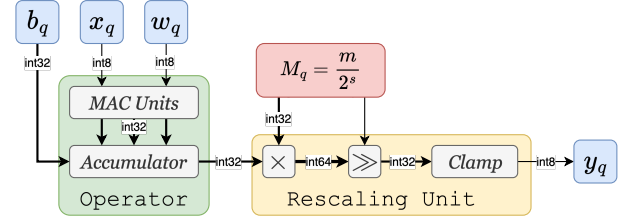


Fig. 1: Visualization of full-integer arithmetic proposed in [1].

A. Related Work: Integer-only arithmetic for efficient AI inference on resource-constrained devices was introduced by Jacob et al. [1], along with Quantization-Aware Training (QAT), enabling full-integer quantization with 8-bit weights and 32-bit rescaling while maintaining accuracy.

Subsequent works refined quantization techniques to improve accuracy and hardware efficiency. Nagel et al. [7] proposed weight equalization and bias correction to reduce quantization-induced bias. Li et al. [3] introduced Additive Powers-of-Two (APoT) weight quantization, enabling shift-and-add based, low-cost dot-product. Efficient rescaling has been addressed by training all quantization thresholds under Power-of-Two (PoT) constraints [8]. This was extended with hardware-compliant schemes with better resolution near zero to improve accuracy [5].

Existing power-of-two methods have advanced the field of integer-only inference, by a fixed trade-off of cost for model quality. A fine-grain cost-quality trade-off in rescaling remains not explored in sufficient detail. This work addresses that gap by proposing *Rescale-Aware Training*, a technique that adapts pre-trained models to low-bitwidth rescaling multiplicands without compromising accuracy.

II. METHODOLOGY

Our implementation is based on the LiteRT framework, which compiles models into a hardware-agnostic intermediate representation. We modify this framework to enable accurate inference modeling under specific rescaler configurations. The same approach is used to implement rescale-aware training.

A. Preliminaries: This section formalizes the quantized inference setting and introduces the relevant assumptions, quantization schemes, and computational models.

1. Quantization Scheme: We assume an integer-only inference pipeline with the following quantization configurations:

- **Weights:** Symmetric, per-channel, 8-bit, scale S_w
- **Activations:** Affine, 8-bit, scale S_x , zero-point Z_x
- **Bias:** Symmetric, per-channel, 32-bit, scale $S_x \cdot S_w$

This format aligns with all widely adopted deployment practices (e.g., LiteRT, TOSA, ExecuTorch) and enables efficient integer-arithmetic hardware.